Chemistry in Everyday Life

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Chemistry in Everyday Life

Durg or Medicine

The essential standards of science impact each circle of human existence. The standards of science have been utilized to support humankind. Consider neatness—the materials like cleansers, family blanches, tooth glues, and so on will ring a bell. Look towards the delightful garments — promptly synthetic substances of the engineered strands utilized for making garments and synthetics offering tones to them will strike a chord. Obviously, affliction and illnesses help us to remember meds — again synthetic compounds. Explosives, energizes, rocket propellants, building and electronic materials, and so on, are for the most part synthetic compounds. Most of the things like explosives ,rocket propellants etc are made up of man made compounds. Our day to day life is influenced by science up to that extent, without realising that we run over synthetic compounds at each second; that we ourselves are wonderful substance manifestations and every one of our exercises are constrained by synthetic substances..

Medications or Drugs

Drugs are substances that interact with a biological system to produce a biological response.

If this response is remedial then these substances are called medicine which are utilized to examine, prevent and cure various ailments.

Use of chemicals for remedial effect is called chemotherapy.

Categorization of Medications

There are four main ways in which drugs might be categorized.

OBJECTIVES:

- To find out the role of chemistry in day to day life
- To understand the mode of categorization of drugs
- To understand the meaning of chemotherapy
- To discuss how drug and target interact
- To understand how different types of drug works
- Learning about food preservatives
- To understand the role of cleansing agents

Points to remember

Activity of Drug decreases with increase in length of carbon chain, because the solubility of substances lowers and they are unable to get into blood and thus donot cause required effect.

Points to remember

Pharmacophore group is responsible for the activity of drug. On removing pharmacophore groups, the drug loses its activity.

(a) By pharmacological effect

This categorization depends on the pharmacological impact of the medications. This is useful for specialists to know the entire scope of drugs available for a specific ailment.

For instance, analgesics have tormented killing impact, cleaning agents kills or obstructs the growth of germs or microorganisms.



(b) By target system

It relies upon the activity of a medication on a specific biochemical interaction. For instance, the activity of compound named "histamine" is restrained by antihistamines. There are different manners by which the activity of histamines can be hindered.

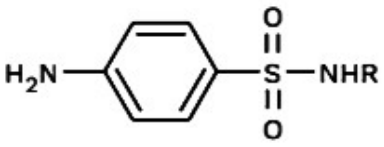


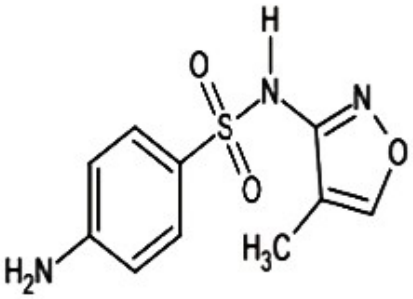
(c) By chemical structure

This categorization relies on the skeleton of the medication .In some cases, biological activity and mechanism of action is the same for the structures involved.

Sulphonamide is an example of this categorization.







Sulfamethoxazole

(d) By target molecule

Some drugs are grouped on the basis of molecular target with which they interact.In many cases, molecular targets are biomolecules like carbs, lipids,proteins and nucleic acid. In some cases, drugs having common skeleton possess same mechanism of action on targets. This categorization is most useful for chemists.

Enzymes

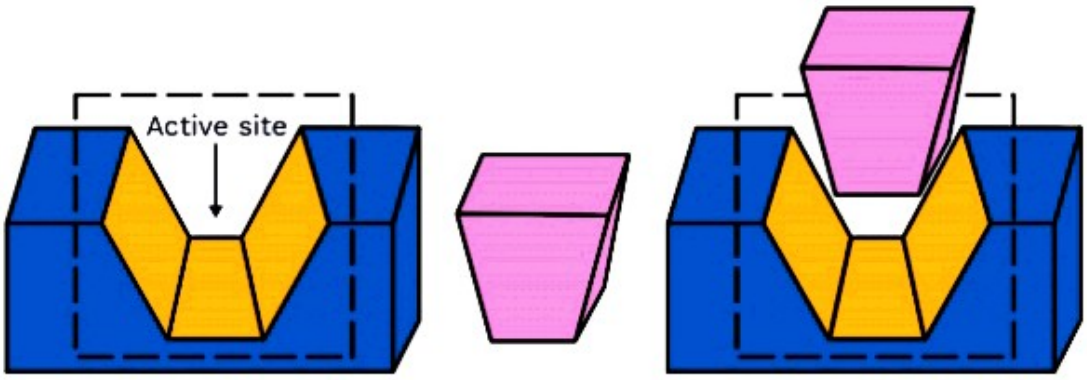
Enzymes are proteins that possess catalytic activity Catalysts speed up chemical reactions. The molecules whereupon enzymes may act are called substrates, and the enzymes convert the substrates into different molecules known as products.

Enzymes as Drug Targets

(a) Catalytic role of enzymes

FIn order to understand drug enzyme interaction ,it is necessary to learn how they catalyse reactions .Enzymes catalyse reactions by performing two important functions:

(i) In first function, enzyme provide a specific surface to substrate to which it can bind .This specific surface is called active site where substrate reacts with reagent.



(a) Enzyme (b) Substrate (c) Enzyme holding substrate

The interactions responsible for binding of substrate to the active sites of enzyme are ionic bonding, van der waals,hydrogen bonding and dipole-dipole interactions.

(ii) The second task of protein or an enzyme is to give functional groups that will strike the substrate and carry out chemical reaction.

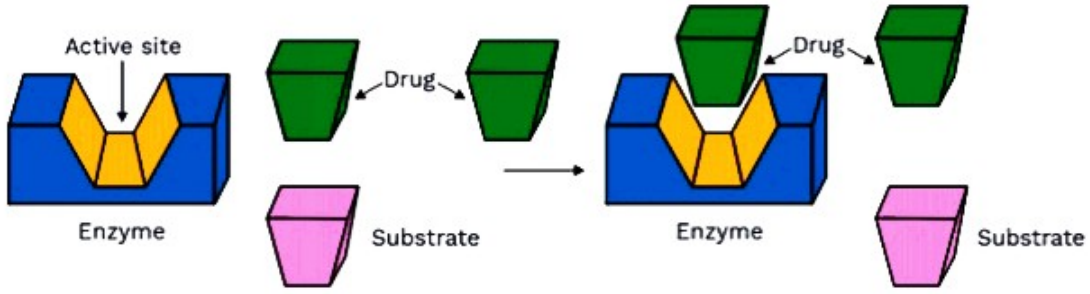
(b) How do drug interact with enzyme?

Drugs help in inhibition of activities of enzyme by blocking the active site of enzyme or

preventing the substrate from binding or hindering enzyme's catalytic activity. Such drugs are termed as **enzyme inhibitors.**

Inhibition of binding of substrate to active site of enzyme is done by drugs in two distinctive ways :

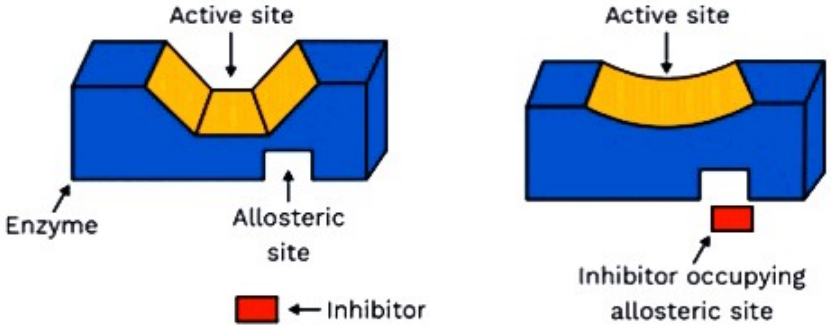
(i) Drugs act as competitive inhibitors means they compete with substrate for binding on specific sites of enzyme.



Drug and substrate competing for active site of enzyme

Drug blocks the active site of enzyme

(ii) Some drugs bind to allosteric sites (i.e., different site of enzyme) which results in change in shape of active site of enzyme. Due to this change in shape of active site, substrate is unable to identify it . If there is covalent bond between enzyme and drug, it is obstructed permanently which results in deterioration of enzyme inhibitor complex and formation of new enzyme takes place.

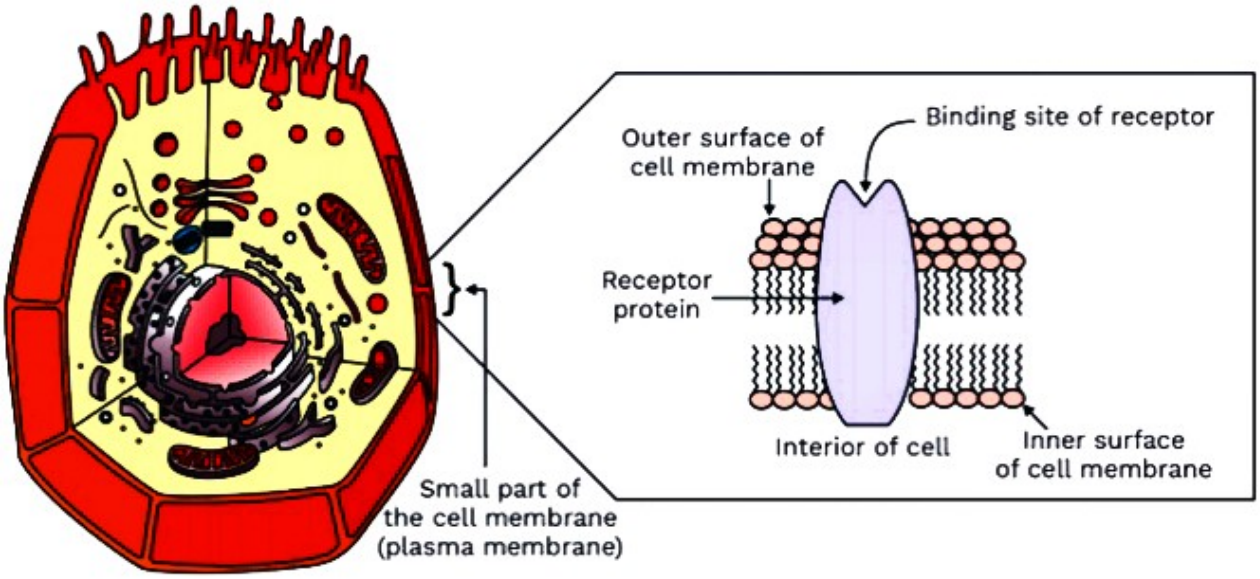


Receptors

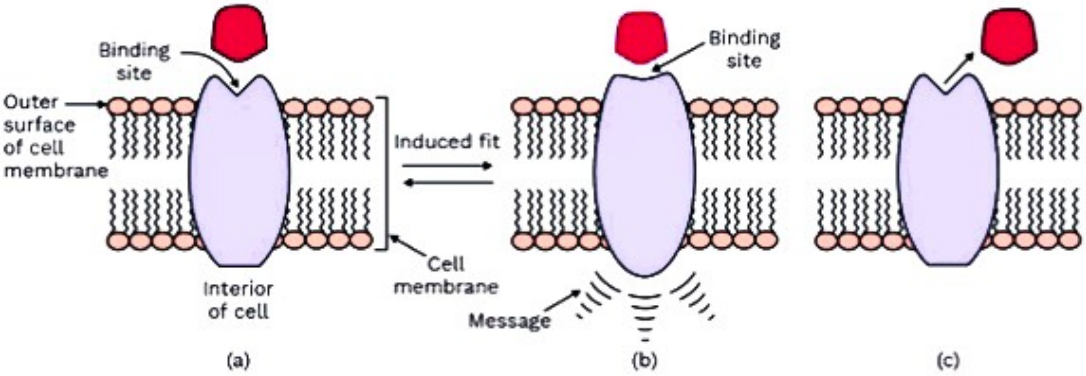
Receptors are proteins that are vital to body's communication process..

Role of receptors

In plasma membrane, receptors are present in a very unique way that their little part having binding site projects out of the surface of plasma membrane and also opens on the outside region of plasma membrane.



Receptor proteins receive chemical messengers at their binding sites. Chemical messengers are the chemical substances which helps in transmitting the message between two neurons and neurons to muscles. Receptor site changes its appearance to accommodate the chemical messenger which results in transfer of message to the cell without entering it.



Different chemical messengers interact with different receptors present in the body.

The binding sites of these receptors have different appearance, structure and amino acid composition. That's why they displays suitable chemical messenger for particular receptor.

Antagonists

Drugs that stick to the binding site of receptor protein and also hinder its function are called **antagonists**. Antagonists are given to obstruct a message.

Agonists

Drugs that imitate the natural messenger and therefore stimulates an action are known as **agonists**. Incase of shortage of chemical messenger in body, agonists are given .

Therapeutic Effects of Drugs and Chemical substances in Food

Therapeutic effects of various kinds of drugs

Antacids

Medicine which neutralizes gastric acid in stomach is known as antacid. They are used in upset stomach, dyspepsia and heart burn.

Mechanism of action

Histamine stimulates the secretion of hydrochloric acid and pepsin in the stomach. Whenever an excess amount of gastric juices is released, the mucous barrier present in stomach deteriorates, leading to irritation and pain. Antacid helps in neutralizing excess gastric acid present in stomach and provide relief from pain.They also reduce damage to the oesophagus and stomach mucous barrier.

Previous Year's Questions

The correct statement(s) among I to III with respect to potassium ions that are abundant within the cell fluids is/are:

(I) They activate many enzymes

(II) They participate in the oxidation of glucose to produce ATP

(III) Along with sodium ions, they are responsible for the transmission of nerve signals

(A) I and II only (B) I and III only

(C) I, II and II (D) III only

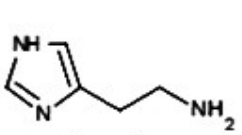
Ans. C



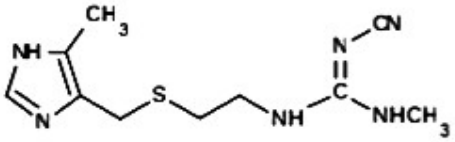


Example :

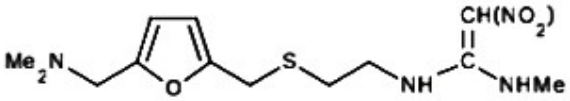
Tagamet (cimetidine) was the first breakthrough in anti ulcer therapy .This medicine forestall the interaction between histamine and the receptors. These receptors are present in the stomach wall .Due to this less amount of acid is released. This drug was the largest selling drug for long time until Zantac (ranitidine), was discovered.



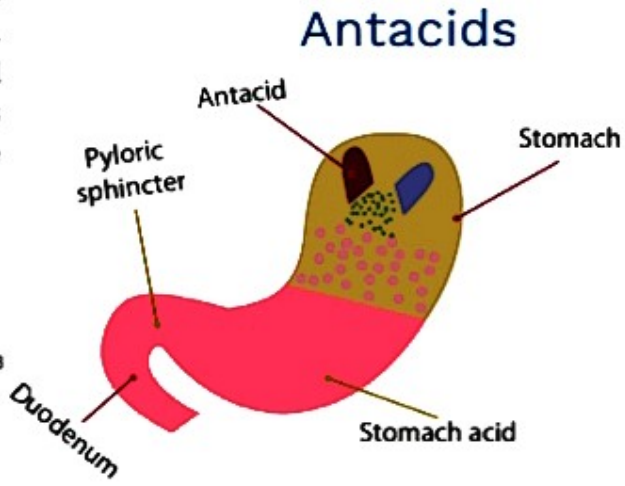
Histamine



Cimetidine



Ranitidine



Antihistamines

Drug that obstructs the physiological impacts of histamine, utilized particularly in the treatment of allergies.

Mechanism of action

Histamine is a powerful vasodilator. It perform various functions in our body. It contracts the smooth muscles in the gut and bronchi and loosens up other muscles. It is responsible for allergic response to pollen and stuffy nose in common cold.Antihistamines compete with histamine for active sites of receptor which results in hinderance in the action of histamine. Thus, provide relief from various allergies.

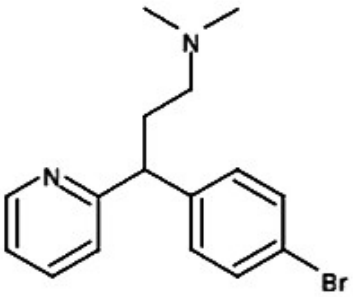
Example: Brompheniramine and terfenadine are examples of antihistamines.

Previous Year's Questions

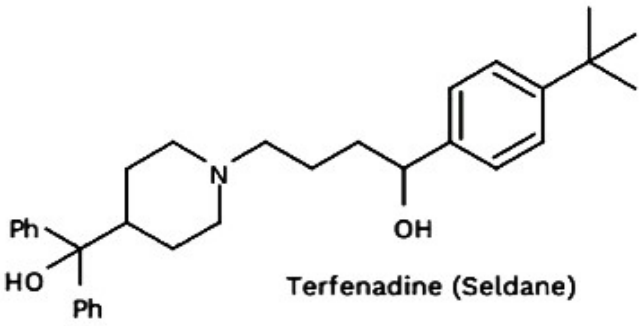
The mechanism of action of Terfenadine (Seldane) is:

(A) Activates the histamine receptor
(B) Helps in the secretion of histamine
(C) Inhibits the secretion of histamine
(D) Inhibits the action of histamine receptor

Ans. D



Brompheniramine
(Dimetapp, Dimetane)



Terfenadine (Seldane)

NOTE: Antiallergic and antacid drugs work on different receptors.

Neurologically Active Drugs

Neurologically active drugs such as tranquilizers and analgesics influence the message transfer mechanism between nerve and receptor.

(a) Tranquilizers

Drug which is used to cure stress, and mental ailments is termed as tranquilizer.

Mechanism of action

Tranquilizers produce sedation through action of central nervous system and also help in muscle relaxation.

Uses:

(i) Tranquilizers induce a sense of being comfortable and happy to provide relief from anxiety ,stress and irritability.

(ii) These affect the central nervous system and induce sleep for the patients.

(iii) The essential ingredient of sleeping pills is tranquilizer.



Previous Year's Questions

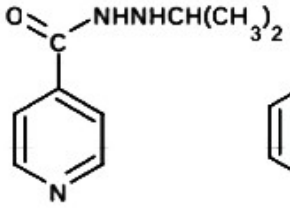
If a person is suffering from the deficiency of nor-adrenaline, what king of drug can be suggested?

(A) Analgesic
(B) Antidepressant
(C) Anti-inflammatory
(D) Antihistamine

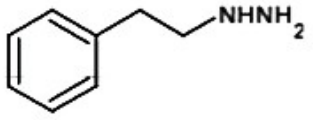
Ans. B

Example

Noradrenaline is an example of neurotransmitters. It helps in mood changes. Low levels of noradrenaline i.e signal-sending activity becomes low can result in depression. In such cases, antidepressants are required. These drugs obstruct the catalysing enzyme of degradation of noradrenaline. If the catalysing enzyme is obstructed, this results in slow metabolism and can activate its receptor for longer periods of time, thus preventing the effect of depression. Phenelzine and iproniazid are examples of antidepressants.

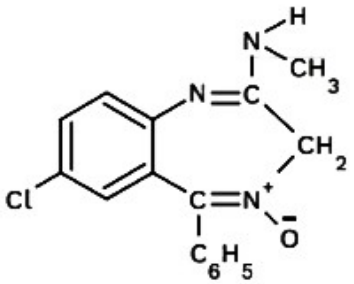


Iproniazid

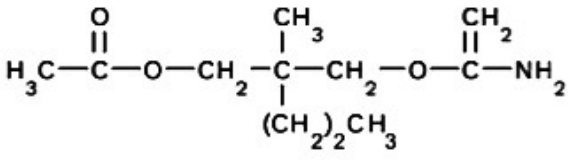


Phenelzine (Nardil)

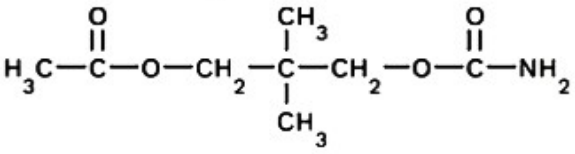
Some tranquilizers namely, meprobamate and chlordiazepoxide, are mild tranquilizers used for relieving tension. For controlling depression and hypertension, equanil is used.



Chlordiazepoxide



Meprobamate



Equanil

Barbiturates: Veronal, luminal, Seconal etc., are sleep producing agents and also termed as barbiturates. Serotonin and Valium are also examples of tranquilizers..

Previous Year's Questions

Noradrenaline is a/an:

(A) Antacid
(B) Neurotransmitter
(C) Antidepressant
(D) Antihistamine

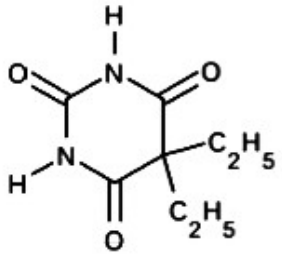
Ans. B

Previous Year's Questions

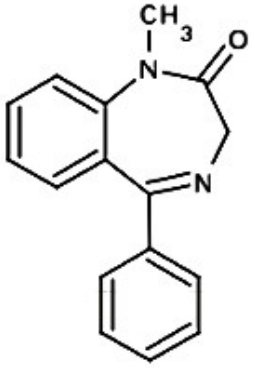
Which one of the following types of drugs reduces fever?

(A) Analgesic
(B) Antipyretic
(C) Antibiotic
(D) Tranquiliser

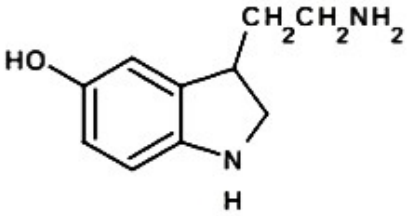
Ans. B



Veronal



Valium



Serotonin



(b) Analgesics

Analgesics abolish or reduce pain without causing mental confusion , impairment of consciousness, paralysis or incoordination etc.

Types of Analgesics

(i) Non-narcotic (non-addictive) analgesics

Aspirin and paracetamol are examples of non-narcotic analgesics.

Example Aspirin

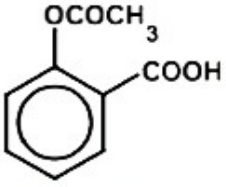
- Aspirin obstructs the synthesis of chemicals known as prostaglandins .Prostaglandins cause inflammation in the tissue which is associated with the symptoms of pain ,swelling ,redness and fever.
- Drugs such as non narcotic analgesics provide relief from skeleton pain.
- These drugs are also used in preventing platelet coagulation and reducing fever (antipyretic).
- Aspirin is also used in prevention of heart attacks.

(ii) Narcotic drugs

When administered in medicinal doses, Morphine and many of its homologues relieve pain and produce sleep. In

Previous Year's Questions

The compound



is used as

(A) antiseptic
(B) antibiotic
(C) analgestic
(D) pesticide

Ans. C

Points to remember

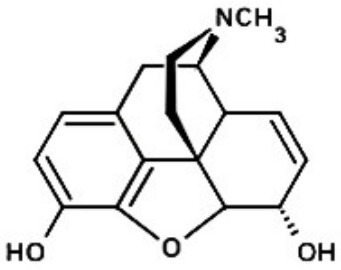
m and p hydroxy benzoic acids are inactive while o- hydroxy benzoic acid is active in aspirin

poisonous doses, these produce coma, stupor, convulsions and finally death. Sometimes morphine narcotics are referred to as opiates, because they are acquired from the opium poppy.

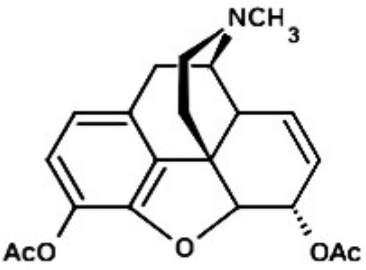
These analgesics are mainly used for the relief of cardiac pain and postoperative pain .They are also used to reduce pains of terminal cancer.



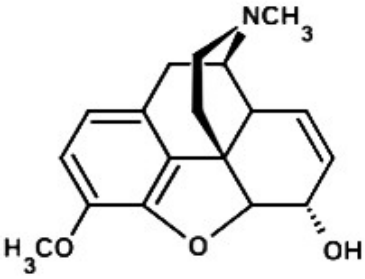




Morphine



Heroin



Codeine

Antimicrobials

Microorganisms like virus, bacteria etc., cause many diseases in living beings.. These drugs tend to prevent / destroy development or obstruct the pathogenic action of microbes such as fungi (antifungal agents), bacteria (antibacterial drugs), parasites (antiparasitic drugs),virus (antiviral agents).

Antibiotics, antiseptics and disinfectants are examples of antimicrobial drugs.

(a) Antibiotics

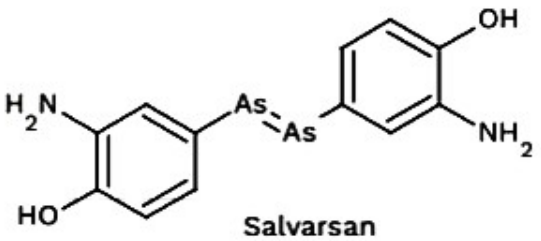
These are used to cure and prevent bacterial infections . Antibiotics have minimal or no toxicity for humans. These drugs hinders the metabolic processes of microorganisms which results in inhibition of growth of microorganisms. An antibiotic also destroys microorganisms.

Examples:

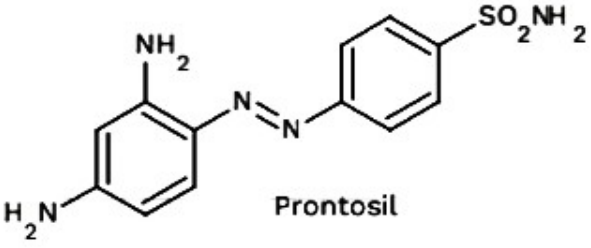
One of the most effective antibiotic is sulphapyridine.



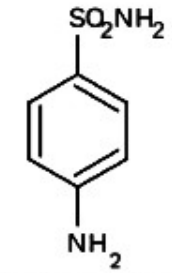
Paul Ehrlich is a German bacteriologist who got Nobel prize for the discovery of medicine named salvarsan in 1908.In order to find cure for syphilis, he investigated arsenic based structures and developed the medicine salvarsan. It is also known as arsphenamine. It was the first effective breakthrough in the treatment of syphilis .Although arsphenamine is very effective on syphilis causing bacteria, spirochete but is also toxic to humans.



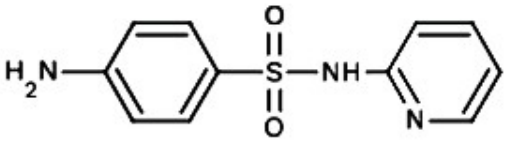
Salvarsan



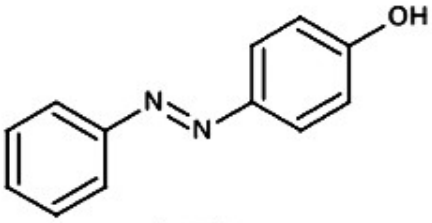
Prontosil



Sulphanilamide



Sulphapyridine



Azodye

Types of Antibiotics :

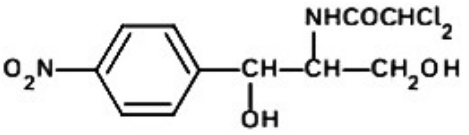
1. **Bactericidal:** Antibiotic that kill bacteria. e.g Penicillin , Aminoglycosides , Ofloxacin etc.
2. **Bacteriostatic:** Antibiotic that obstructs the growth of bacteria. e.g Erythromycin , Tetracycline , Chloramphenicol etc.

Classification based on Spectrum of action

Antibiotic affects a range of microorganisms and that range is its spectrum of action.

(i) Broad spectrum antibiotics

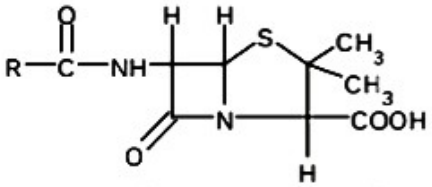
Broad spectrum antibiotic is an antibiotic that kill or obstructs Gram-positive and Gram-negative bacteria .e.g. Vancomycin , Ofloxacin and Chloramphenicol .Chloramphenicol drug can be taken orally in diseases like dysentery, typhoid, acute fever, certain form of urinary infections, pneumonia and meningitis as it is quickly absorbed in gastrointestinal tract.



Chloramphenicol

(ii) Narrow spectrum antibiotics

Antibiotics which are effective against Gram-negative or Gram-positive bacteria are called narrow spectrum antibiotic. e.g., Penicillin Limited spectrum antibiotics are antibiotics which works effectively on single organism or ailment.



General Structure of Pencillin

(b) Antiseptics and disinfectants

Chemical substances that are used to control infections by killing or forestalling the development of microorganisms.

Antiseptics

Chemical substances which are applied to the external surface of body like injuries, cuts, diseased skin surfaces and ulcers. Hydrogen peroxide, furacine, chloroxylenol , soframycine, etc. are examples of antiseptics. These are not consumed like antibiotics.

Commonly used antiseptic:

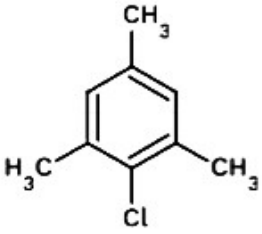
(i) Dettol contains terpineol and chloroxylenol.

(ii) Bithionol is an additive used in soaps for its antiseptic properties.

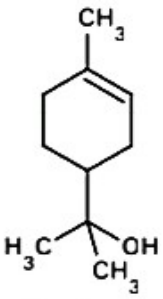
(iii) Tincture of iodine consists of 2-7 % elemental iodine in alcohol - water. Iodine is applied on injuries because of its antiseptic properties. .

(iv) Iodoform is also used to control infections on wounds.

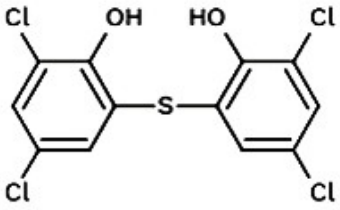
(v) Boric acid eye wash is used to clean eyes as it is a weak antiseptic.



Chloroxylenol



Terpineol



Bithionol

Disinfectants

Chemical substances which are used to inactivate or destroy microorganisms on inert surfaces are known as disinfectants.

Uses:

(i) They are used on floors, instruments, etc.

(ii) They even destroy microorganisms present on the skin.

(iii) They are used in hospitals, , kitchens etc., to disinfect them.

(iv) Few disinfectants work as antiseptic as well as disinfectant only by changing their concentration.

Example:

1 % solution of phenol , Chlorine bleach, sulphur dioxide (very low concentrations), are examples of disinfectants.

Points to remember

Substituted compounds are generally toxic than parent hydrocarbon. C_6H_5OH is highly toxic and a stronger antiseptic than C_6H_6. Salicylic acid is a strong antifungal and antirheumatic agent than the benzoic acid.

Antifertility Drugs

Drug which are taken to control unwanted pregnancy is known as antifertility drug.

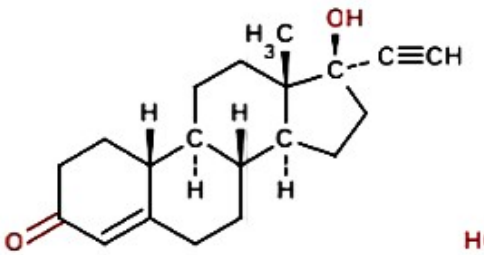
These drugs contain a combination of progesterone derivatives and synthetic oestrogen.

Progesterone prevents pregnancy by suppressing ovulation. Synthetic progesterone derivatives are very effective.

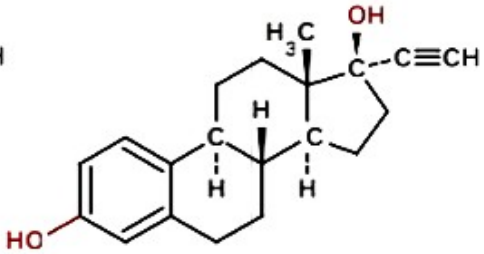


Example

Synthetic progesterone derivative such as Norethindrone used as oral contraceptives. The combination of estrogen and progesterone derivative also known as novestrol (Ethynylestradiol) is used as antifertility drug.



Norethindrone



Ethynylestradiol (Novestrol)

Food Additives

Chemical substances are added in food materials for

(i) Preserving them,

(ii) Enhancing their appeal,

(iii) Adding nutritional value.

Previous Year's Questions

The antifertility drug Novestrol can react with

(A) Br_2/water; $ZnCl_2$/HCl; $FeCl_3$

(B) $ZnCl_2$/HCl; $FeCl_3$; Alcoholic HCN

(C) Br_2/water; $ZnCl_2$/HCl; NaOCl

(D) Alcoholic HCN; NaOCl; $ZnCl_2$/HCl

Ans. A

Main groups of food additives are as follows:

(1) Food colours
(2) Flavours and sweeteners
(3) Fat emulsifiers as well as stabilising agents
(4) Flour improvers
(5) Antioxidants
(6) Preservatives
(7) Nutritional supplements like minerals, amino acids etc.

Chemicals of category (7) have nutritional value. Unlike category 7,chemicals of category 1 to 6 do not possess nutritional value but they are used for increasing life of stored food. Sometimes, they are also used for cosmetic purposes.

Artificial Sweeteners

Artificial sweeteners, also known as synthetic sugar substitute. Natural sweeteners, e.g., honey increase calorie count in our body .That's why everyone prefer artificial over natural sweetener.

Example:

Ortho-sulphobenzimide
is approximately 550 times sweeter as compared to natural sweetener and also discharged from our body in unaltered form.
is harmless and very useful for those who wants to control their calorie intake and patients suffering from diabetes.

Previous Year's Questions

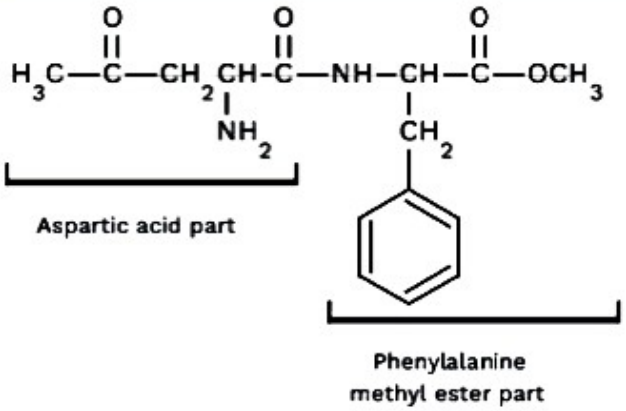
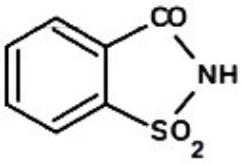
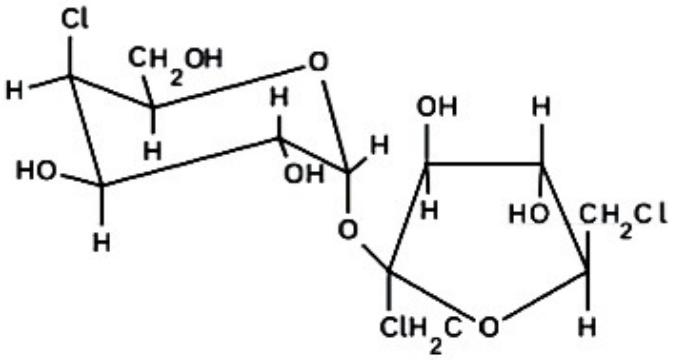
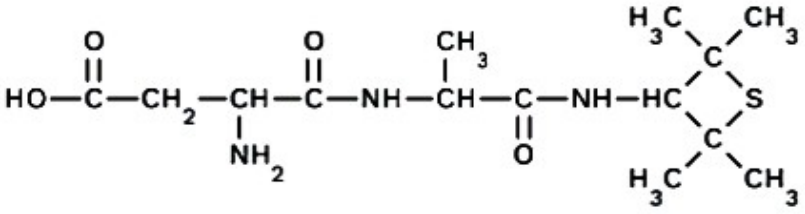
Which artificial sweetener contains chlorine?

(A) Aspartame
(B) Saccharin
(C) Sucralose
(D) Alitame

Ans. C



Mostly used artificial sweeteners are as follows:.

Artificial sweetening agents	Structure	Value of sweeteness
Nutra sweet or Aspartame		100
Saccharin		550
Sucralose		600
Alitame		2000

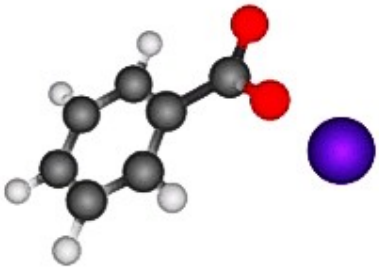
Aspartame also known as nutra sweet is approximately 100 times sweeter as compared to natural sweetener.
Nutra sweet is only used in cold food and cold drinks because of its unstability at cooking temperature.
Alitame is high intensity sweetener, despite the fact it is more steady than aspartame, It is difficult to control the sweetness of food varieties When it is used.
Sucralose is the one which completely taste like sugar and also its appearance is similar. But, it does not provide calories like sugar and also stable at cooking temperature.

Role of Preservatives in Food

Preservatives in food are used to prevent spoilage of food due to growth of microbes . Sugar, table salt, sodium benzoate etc., are most common preservatives used in food.







Sodium benzoate is used in food to prevent its spoilage by fungi but in limited amount. It is metabolised in human body. Propanoic acid and salts of sorbic acid are also examples of food preservatives.

Role of Antioxidants in Food

Antioxidants are very important food additives as they prevent rancidity and increase shelf life of food.They hinder the action of oxygen on fat containing food varieties because they are more reactive towards oxygen as compared to food material.

Example:

The two most recognizable antioxidants are butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT). In order to increase the storage time of butter, Butylated hydroxyanisole is added to butter.
Sometimes combination of BHT ,BHA and citric acid are added to food materials as they are more effective. Antioxidants useful for cut, peeled or dried fruits and vegetables ,wine and beer, sugar syrups etc. are sulphite and sulphur dioxide.

Cleansing Agents

Cleansing agents are substances which are used to remove dirt, including dust, stains, bad smells, etc. by removing fats that bind other materials to the skin or fabric. These also play an important role in enhancing the cleansing properties of water.

Types of Cleansing agent

(a) soaps



(b) Synthetic detergents.

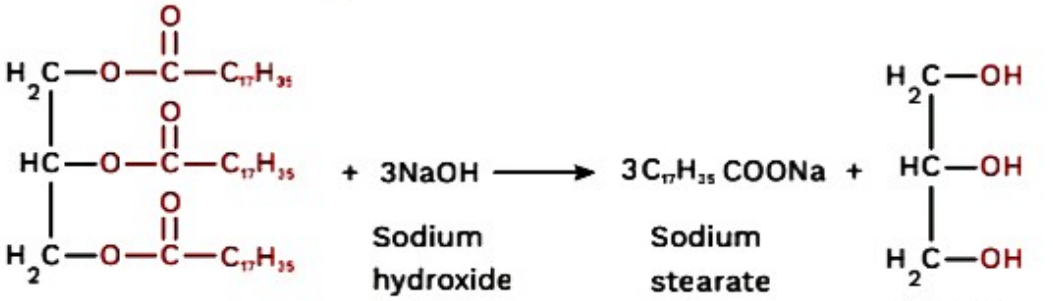


Soaps

Soaps are the cleansers utilized since long. They are sodium or potassium salts of long chain fatty acids, e.g., stearic, palmitic acids etc.

Saponification.

In saponification ,fat is heated (i.e., glyceryl ester of fatty acid) with aqueous sodium hydroxide solution to form soaps containing sodium salts.



Glyceryl ester or stearic acid (Fat)

Sodium hydroxide

Sodium stearate

Glycerol (or Glycerine)

In saponification, hydrolysis of esters of fatty acids takes place. But the soap prepared remains in colloidal form which is then precipitated out by adding NaCl.

After removing soap from solution ,the remaining solution contain glycerol . Fractional

distillation is used to recover the glycerol.
Potassium and sodium soaps are utilized for cleaning purposes because of their solubility in water. Generally, potassium soaps are softer for skin as compared to sodium cleansers. If KOH solution is used in place of NaOH, potassium soaps are obtained.

Types of Soaps

The basic procedure for soap making remains same for all kinds of soaps. Different types of soaps are prepared by varying their raw materials.

1. **Toilet soaps :**
   Fats and oils of better quality are used to make toilet soaps. Toilet soap making also involves elimination of extra alkali . These soaps are very alluring as beautiful colours and aromas are added while soap making.
2. **Transparent soaps :**
   Soaps that buoy in water are formed by dissolving the soap in ethanol and then evaporating the excess solvent. Their making involves beating air bubbles prior to their hardening.
3. **Medicated soaps :**
   In order to prepare medicated soaps, substances having restorative value are added. Deodorants are also used in some soaps to make it smell nice.
4. **Shaving soaps :**
   They contain glycerol to forestall fast drying. The making of shaving soaps involves addition of rosin to form sodium rosinate which lathers well.
5. **Laundry soaps :**
   In these soaps, fillers are used. Borax , sodium silicate etc., aresome common fillers. Laundry soaps are mainly used for washing clothes.
6. **Soap chips :**
   Soap flakes are tiny broken pieces of soap which are formed from melted cleanser. Soap granules are dried miniature soap bubbles.
7. **Scouring soaps and Soap powders :**
   Soap powders and scouring soaps are comprise of cleanser, a scouring agent and builders. Powdered pumice and finely divided sand are some common scouring agents .

Previous Year's Questions

The distillation technique most suited for separating glycerol from spent-lye in the soap industry is:
(A) Distillation under reduced pressure
(B) Simple distillation
(C) Fractional distillation
(D) Steam distillation
Ans. A

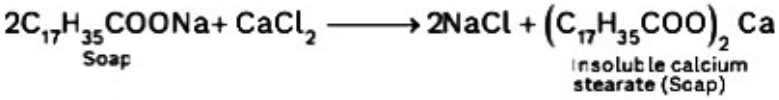
Previous Year's Questions

If you spill a chemical toilet cleaning liquid on your hand, your first aid would be
(A) aqueous NaOH
(B) aqueous $NaHCO_3$
(C) aqueousNH_3
(D) vinegar
Ans. D

Trisodium phosphate and sodium carbonate are the builders which helps the cleansers to act rapidly.

Why soap does not get along with hard water?

Hard water contains a high concentration of bivalent ions such as magnesium and calcium ions. When potassium or sodium soaps are added in hard water, bivalent ions like Ca^{2+} and Mg^{2+} displace K^+ or Na+ ions to form insoluble calcium and magnesium soaps.



Insoluble magnesium and calcium soaps segregate as curdy white precipitates ,also known as scum. These curdy white precipitates sticks to the fibre of the clothes. If hard water is used in washing hairs , hairs looks dull. Even dye doesn't retain equally on fabric washed with soap utilising hard water, due to this gummy mass.

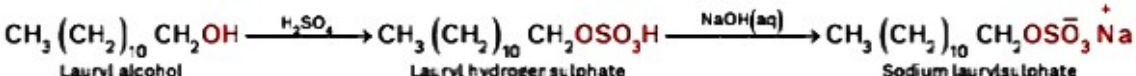
Synthetic Detergents

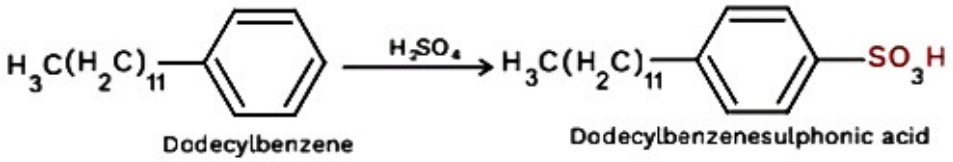
Synthetic detergents are chemical substances which shows properties of soaps but do not contain any soap. They work in both soft and hard water .Unlike soaps, froth is produced by detergents in hard water.Some detergents also give froth in super cold water.

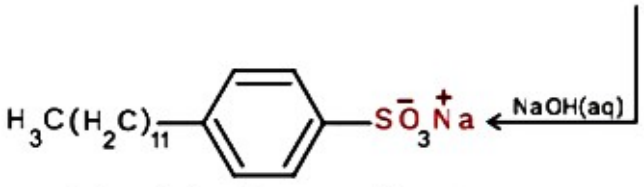
Synthetic detergents are mainly categorized into three categories:

(i) Anionic detergents
(ii) Cationic detergents
(iii) Non-ionic detergents

(i) **Anionic Detergents:** Sodium salts of sulphonated long chain hydrocarbons or alcohols are called anionic detergents. Alkyl hydrogen sulphates prepared by reacting long chain alcohols with concentrated H_2SO_4 are neutralised with alkali(NaOH) to produce anionic detergents. Similarly, In the same way, alkyl benzene sulphonates are obtained by reacting alkyl benzene sulphonic acids with alkali (NaOH).







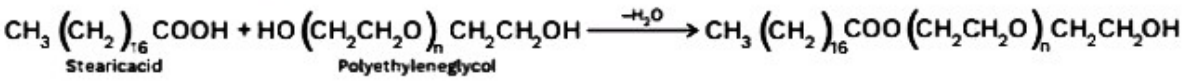
In anionic detergents, the anionic part (lipophilic hydrocarbon group) of the molecule is engaged in the cleansing work. They are used in toothpastes and also for household work.

Example: Sodium salts of alkyl benzene sulphonates.

(ii) Cationic Detergents: These detergents are quarternary ammonium salts of amines with bromides or chlorides ,acetates are called as cationic detergents. A long hydrocarbon chain and a positive charge on N atom comes in cationic part. These detergents are for limited use because they are costly and also possess germicidal properties.

Example: Cetyltrimethylammonium bromide.

(iii) Non-ionic Detergents: Non-ionic detergent is prepared by reacting stearic acid with polyethylene glycol.There are no ions present in non-ionic detergent's constitution.



For liquid dishwashing, non-ionic detergents are used. Both soap and non ionic detergents have same mechanism of cleansing action .Non ionic detergents also form micelle for removing oil and grease.

In case of highly branched hydrocarbon chain of detergent , it is impossible for bacteria to decompose it .This is the biggest demerit in its utilization. The detergents are accumulated due to their slow deterioration. Water of Rivers, ponds etc., are contaminated due to effluents. Even after sewage treatment ,these effluents containing such detergents stays in water and making it unfit for human use.

Nowadays, To prevent pollution, hydrocarbon chain is less branched as they are biodegraded easily

Previous Year's Questions

Thiol group is present in:
(A) Methionine
(B) Cytosine
(C) Cystine
(D) Cysteine
Ans. D

Summary



- Chemical that affects our metabolism and provides treatment from ailments is called drug or medicine. Never take them in excess or without doctor's prescription as they are harmful.
- Medications mainly target proteins ,lipids carbohydrates etc.,(biological macromolecule known as target molecules.
- Medications interact with explicit targets which limits the side effects .It also confines drug's action. Thus, drugs like antibiotics, tranquilizers, antiseptics, analgesics, antacids and disinfectants are utilized for specific purpose. Antifertility drugs are important to control increasing population.
- Food additives such as preservatives, antioxidants, flavours, nutritional supplements and edible colours are added to the food to make it alluring, tasteful. Surprisingly, they also have nutritive value. Preservatives in food forestall spoilage due to growth of microbes. Artificial sweeteners are vey important for the ones who want to control their calorie consumption and also patients who are not allowed to eat anything sweet such as diabetic patients
- Detergents are preferred over soaps as they get along well with hard water. Non-ionic, cationic and anionic are three types of detergents.
- Straight chain hydrocarbon detergents are preferred as they are biodegradable. Detergents with branched chains are non- biodegrradable consequently cause environmental pollution.

Q1 **Among the following sets ,which one contains only medicinal compounds is**

(A) boric acid, morphine, 9-oxodecanoic acid

(B) Phenacetin, morphine Alizarin,

(C) Aspirin, Boric acid, chloramphenicol,

(D) Phenolphthalein, Aspirin, Gention violet

Ans. (C)

Aspirin is an analgesic as well as antipyretic. Boric acid is mild antiseptic. chloramphenicol is an broad spectrum antibiotic .

Q2 **Which is an example of broad spectrum antibiotic?**

(A) Chloramphenicol **(B) Xylocane**

(C) Antiseptic **(D) Plasmoquin**

Ans. (A)

Broad spectrum antibiotic is an antibiotic that kill or obstructs Gram-positive and Gram-negative bacteria, eg, chloroamphenicol is quickly absorbed from the gastrointestinal tract and can be given orally in case of typhoid, dysentry, Acute fever, certain form of urinary infections, meningitis and pneumonia.

Q3 **Which of the following statement is not true about the drug barbital?**

(A) It causes addiction **(B) It is a non-hypnotic drug**

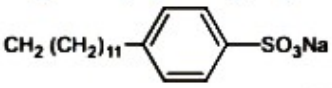
(C) It is tranquillizer **(D) It is used insleeping pills**

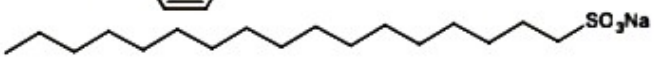
Ans. (B)

Barbital or varonal is a sleep producing, ie, hypnotic tranquillizer. It is used in sleeping pills and cause addiction.

Q4 **Which detergent can cause maximum pollution?**

(A) $CH_3CH(C_4H_9)$-$CH_2CH_2CH(C_4H_9)$ $(CH_2)_3CH(C_4H_9)CH_2CH_2SO_3Na$

(B) 

(C) 

(D) Detergents are always pollutio

Ans.(A)

Greater the branching, greater is the non-biodegradability of a detergent. A non-biodegradable detergent cause maximum pollution.

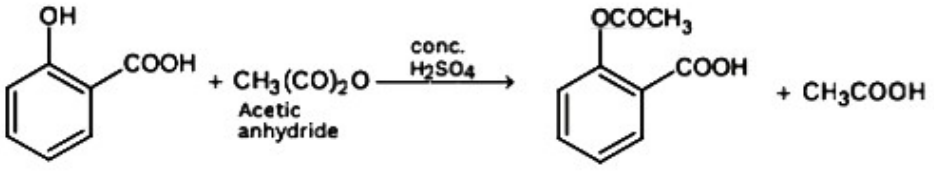
Q5 **The reagent (s) used in the preparation of aspirin from salicylic acid**

(A) $SOCl_2$, **pyridine** **(B)** $(CH_3CO)_2O,H$

(C) CH_3CO_2H,HCl **(D)** $CH_3Cl,AlCl_3$

Ans.(B)

Aspirin (2-acetoxy benzoic acid) is an analgesic. It is obtained by acetylating salicylic acid with acetic anhydride and conc H_2SO_4



salicylic acid asprin

Q6 **Another name for Tranquiluisers is**

(A) Psychosomatic drugs **(B) Psychotherapeutic drugs**

(C) Psychosystolic drugs **(D) None of the above**

Ans. (B)

Tranquilisers are the drugs which used to relieve mental disorders. Tranquilisers are sometime called as psychotherapeutic drugs as they act on the central nervous system.

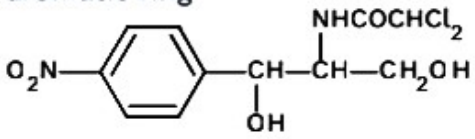
Q7 **Which of the following antibiotic contains nitro group attached to aromatic nucleus in its structure ?**

(A) Pencillin **(B) Streptomycin**

(C) Tetracyclin **(D) Chloramphenicol**

Ans.(D)

Among the given antibiotics, only chloroamhenicol contains a nitro group attached to aromatic ring



Chloramphenicol

Q8 **Morphine on alkylation and acetylation gives, respectively**

(A) Heroin, codeine **(B) Heroine, meperidine**

(C) Codeine, heroin **(D) Meperidine, heroin**

Ans. (C)

Morphine on alkylation and acetylation gives Codeine, heroin respectively.

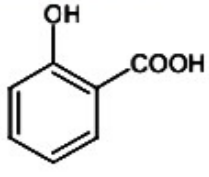
Q9 **The functional groups present in 'salol' are**

(A) $-NH_2$ and $-OR$ **(B) $-OH$ and $-COR$**

(C) $-NH_2$ and $-COOH$ **(D) $-OH$ and $-COOR$**

Ans. (D)

The functional group present in salol are —OH and —COOR. Salol is phenyl salicylate. Its structure is as



salol

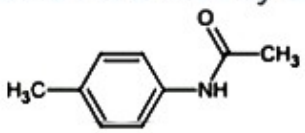
Q10 **A drug that is antipyretic as well as analgesic is**

(A) Chloroquine **(B) Penicillin**

(C) Paracetamol **(D) Chloropromazine hydrochloride**

Ans. (C)

Analgesic drugs are used in relieving pain while antipyretic drugs are used to control fever. Paracetamol is wideley used as an antipyretic as well as analgesic.



Paracetamol
(4-hydroxyacetaniide)

Q11 **Disinfectants are different from antiseptics as**

(A) Disinfectant kill the microorganism and antiseptic merely inhibit the growth.

(B) Disinfectants are used against insects while antiseptic are used against microorganism .

(C) Disinfectants can be taken orally also while antiseptic are used only over skin.

(D) Disinfectants cannot be used over living tissues while antiseptic are used over living tissues.

Ans.(D)

Antiseptics and disinfectants both inhibit and kill the microorganisms but disinfectants are not safe to apply over living tissues.

Q12 **The detergent which possess germicidal properties is**

(A) Sodium laurylsulphate

(B) Cetyltrimethyl ammonium chloride

(C) Lauryl alcohol ethoxylate

(D) Sodium-2-dodecylbenzenesulphonate

Ans. (B)

Cationic detergents are utilized as germicides. Cationic detergents are usually quaternary ammonium salts. Thus, cetyltrimethyl ammonium chloride, a quaternary ammonium salt is a cationic detergent and used as a germicide.

Q13 **Which of the following is not a artificial sweetener?**

(A) Sucralose **(B) Alitame**

(C) Saccharin **(D) Sucrose**

Ans. (D)

Saccharin is 110 times sweeter than sugar. Aspartame is 180 times sweeter than sugar Sucralose is 650 times sweeter than sugar.

Q14 **Which is correct about vanillin?**

(A) A flavouring agent having vanila flavour

(B) o-hydroxy-m-methoxy benzaldehyde

(C) A food additive

(D) All of the above

Ans.(D)

All are characteristics of vanillin.

Q15 **Chloromycetin (Chloramphenicol) is effective in the treatment of:**

(A) Tuberculosis **(B) Malaria** **(C) Typhoid** **(D) Cholera**

Ans. (C)

Chloramphenicol is broad spectrum drug, used in treatment of typhoid, dysentery, acute fever, meningitis pneumonia and certain forms of urine infection.

Q16 **One of the most known antiseptic, dettol is a mixture of terpineol and**

(A) Bithional **(B) Chloroxylenol**

(C) o-cresol **(D) Serotonin**

Ans. (B)

Dettol (antisepti(C) is a mixture of 4.8% chloroxylenol +9.9% terpineol and absolute alcohol.

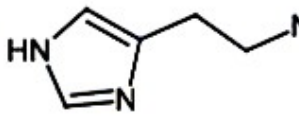
Q17 **Chloramine-T is a**

(A) Disinfectant **(B) Antiseptic**

(C) Analgesic **(D) Antipyretics**

Ans. (B)

Antiseptics drugs cause destruction of microorganisms, which produce, septic diseases, e.g., dettol, savlon, acriflavin, boric acid, phenol, iodoform, $KMnO_4$ and some dyes such as chloramines-T, methylene blue.

Q18 **The drug**  **is used as**

(A) Vasodilator **(B)Analgesics**

(C) Antacid **(D) Antiseptic**

Ans. (A)

The given structure is of histamine that acts as a neurotransmitter and vasodilator.

Q19 **Various phenol derivatives, tincture of iodine (2-3% I_2 in water/alcohol) and some dyes like methylene blue are:**

(A) Antiseptics **(B) Disinfectants** **(C) Analgesics** **(D) Antipyretics**

Ans. (A)

Antiseptic drugs cause destruction of microorganisms that produce septic diseases, e.g., dettol, savlon, acriflavin, boric acid, phenol, iodoform, $KMnO_4$ and some dyes such as chloramine-T, methylene blue, genatian violet.

Q20 **The safest and the most common alternative of sugar is**

(A) Glucose **(B) Aspartame** **(C) Saccharin** **(D) cyclodextrin**

Ans. (B)

The safest and the most common alternative of sugar is aspartame.

Q21 **Which of the following is known as invert soap?**

(A) Pentaerythritol monostearate **(B) Sodium stearyl sulphate**

(C) Trimethyl stearyl ammonium bromide **(D) Ethoxylated nonyphenol**

Ans. (C)

Trimethyl stearyl ammonium bromide is known as invert soap.

Q22 **Trade name of aspartame is**

(A) Alitame **(B) Saccharin** **(C) Sucralose** **(D) Nutra sweet**

Ans.(D)

Nutra sweet

Q23 **Among the following ,which detergent is used to prepare cosmetics.**

(A) DDBS **(B) Polyethylene glycol**

(C) Cetyltrimethylammonium chloride **(D) LAS**

Ans.(C)

Cetyltrimethyl ammonium chloride is a cationic detergent. It is used to prepare cosmetics because it possesses germicidal properties.

Q24 **Which of the following is an example of food antioxidant?**

(A) BTX **(B) BHT** **(C) BHC** **(D) All the three**

Ans.(B)

BHT is used as an antioxidant is food

Q25 **Amoxycillin is semi-synthetic modification of**

(A) Penicillin **(B) Streptomycin**

(C) Tetracycline **(D) Chloramphenicol**

Ans.(A)

Amoxycillin is semi-synthetic modification of Penicillin.

Q26 Which of the following will not enhance nutritional value of food?

(A) Minerals
(B) Artificial sweeteners
(C) Vitamins
(D) Aminoacids

Ans. (B)

Artificial sweeteners

Q27 Which of the following is not a target molecule for drug function in body ?

(A) Vitamins
(B) Proteins
(C) Lipids
(D) Nucleic Acids

Ans. (A)

Vitamins

Q28 Drug which prevent the formation of acid in stomach is :

(A) Omeprazole
(B) Sulphathiazole
(C) Phenylbutazone
(D) Luminal

Ans. (A)

Omeprazole

Q29 Drug which helps to reduce anxiety and brings about calmness is

(A) Tranquillizer
(B) Diuretic
(C) Analgesic
(D) Antihistamine

Ans. (A)

Tranquillizer